CSN305 - Software Defined Networks

Fall 2026 Syllabus, Section NBB, Class Nbr 5197

Course Description

In this project-based course students will examine and then use the functions and components of Software Defined Networks (SDN) to create a network. They will identify challenges when converting static networks to software defined ones. And will be able to analyze the performance of an SDN using verification and troubleshooting techniques to guarantee quality of service.

Prerequisite(s)

CSN205

Instructor Information

Lisa Li - Primary Instructor
Email: lisa.li2@senecapolytechnic.ca

In-Person Mode

This class will be taught via an **in-person** instruction mode. Students are required to attend all classes on campus.

Times and Location

Th 12:35pm-4:10pm in Newnham Bldg K - K1270

Course Learning Outcomes

Upon successful completion of this course the student will be able to:

1. Identify challenges within static networks to effectively propose SDN solutions.
2. Utilize basic concepts of SDN architecture required to implement a simple software-defined network.
3. Build a virtual network environment for testing and evaluating various SDN configurations and deployments.
4. Explain OpenFlow protocol functions to generate policy-driven data flows and traffic engineering to control and prioritize network traffic based on a business scenario.
5. Create customized flows and evaluate their impact on network performance to improve overall network efficiency.
6. Use NFV and SDN technologies to adjust the network to changing requirements.
7. Explain SDN solutions for network automation and future data centers to effectively manage changes in complex networks.

Essential Employability Skills

- respond to written, spoken, or visual messages in a manner that ensures effective communication
- execute mathematical operations accurately
- apply a systematic approach to solve problems
- use a variety of thinking skills to anticipate and solve problems
- locate, select, organize, and document information using appropriate technology and information systems
- analyse, evaluate, and apply relevant information from a variety of sources
- show respect for the diverse opinions, values, belief systems, and contributions of others
- interact with others in groups or teams in ways that contribute to effective working relationships and the achievement of goals
- manage the use of time and other resources to complete projects
- take responsibility for one's own actions, decisions, and consequences
- communicate clearly, concisely, and correctly in the written, spoken, and visual form that fulfils the purpose and meets the needs of the audience

Course Topics

CSN305

- Virtualization and Hypervisors
- Cloud Computing Fundamentals
- Network Function Virtualization (NFV)
- SDN Architecture and OpenFlow
- SDN Control and Data Planes
- SDN Applications and Traffic Monitoring
- Network Visibility and Quality of Experience (QoE)
- SDN Security

Prescribed Texts

Group	Title	Author	ISBN
Required	SDN and NFV Simplified: A Visual Guide to Understanding Software Defined Networks and Network Function Virtualization	Doherty,Jim	0-13-4307399

To find out the cost of books and learning material go here (https://www.bkstr.com/senecastore/shop/textbooks-and-course-materials).

Any courses not listed on the bookstore webpage do not require any resources for purchase. All resources will be provided by your instructor.

Reference Material

All reference material is provided weekly by the instructor via the Blackboard course platform.

Required Supplies

Solid-state drive (SSD) with a minimum available capacity of 500 GB.

Modes of Evaluation

Assessment Type	Percentage
Labs (10) (4% each)	40
Tests (4) (15% each)	60
Total	100%

Note

To obtain a credit in this course, a student must:

- Achieve a grade of 50% or higher in the overall course
- Achieve a weighted average of 50% or higher on all tests
- Satisfactorily complete all labs

Schedule of Topics and Assignments

Day	Date	Agenda/Topic	Reading(s)	Due
Thu	9/10	Virtualization-Hypervisors Lab 1	Week 1 PPT	
Thu	9/17	Cloud Computing Fundamentals Lab 2	Week 2 PPT	Lab 1 (4%)

Thu	9/24	NFV-Core Networking Functionality Lab 3	Week 3 PPT	Lab 2 (4%)
Thu	10/1	SDN architecture OpenFlow Lab 4	Week 4 PPT OpenFlow switch specification	Lab 3 (4%) Test 1 (15%)
Thu	10/8	OpenFlow-related protocols Lab 5	Week 5 PPT OpenFlow switch specification	Lab 4 (4%)
Thu	10/15	SDN Control plane - Network Controllers Lab 6	Week 6 PPT OpenFlow switch specification	Lab 5 (4%)
Thu	10/22	SDN Data plane - Open hardware	Week 7 PPT	Lab 6 (4%) Test 2 (15%)
Thu	10/29	Study Week		
Thu	11/5	SDN application plane Lab 7	Week 8 PPT	
Thu	11/12	Application use cases Lab 8	Week 9 PPT	Lab 7 (4%)
Thu	11/19	Traffic monitoring Lab 9	Week 10 PPT	Lab 8 (4%) Test 3 (15%)
Thu	11/26	Network Visibility-QoE	Week 11 PPT	
Thu	12/3	How SDN & NFV will affect you SDN Security Lab 10	Week 12 PPT	Lab 9 (4%)
Thu	12/10		All chapters	Test 4 (15%) Lab 10 (4%)

Missed Tests/Late Assessments

Due dates for all evaluations and assessments are posted. Evaluations can include projects, podcasts, videos, assignments, quizzes and/or tests and exams. Students are expected to meet the specified dates and deadlines. It is a best practice for all students to keep a copy of all submitted assignments.

Students who have extenuating circumstances that result in their being unable to meet the stated deadline are encouraged to contact their professor(s). A professor may (or may not) grant an extension to a posted due date. Such extension requests must be discussed prior to the due date, or very closely following. Late submission or completion of any assessments may be subject to a penalty grade deduction. Once feedback is posted and/or discussion of the assessment has taken place, students may not submit that version of the assessment for grading.

Feedback on Assessments

Feedback to students regarding graded assessments can be provided in any of the following ways: posted on LEARN@Seneca, added to Grade Centre comments, taken up synchronously, and/or discussed with students.

Students are welcome to discuss feedback on completed and submitted assessments with their professor during a synchronous class, during posted "virtual" office hours, or by a mutually agreed upon appointment.

Student Progression and Promotion Policy

Letter Grade	Percentage Grade
A+	90% to 100%
A	80% to 89%
B+	75% to 79%
B	70% to 74%
C+	65% to 69%
C	60% to 64%
D+	55% to 59%
D	50% to 54%

F	0% to 49% (Not a Pass)
OR	
EXC	Excellent
SAT	Satisfactory
UNSAT	Unsatisfactory

Listed below are a number of important links to Seneca Polytechnic policies.

- Student Progression and Promotion Policy (http://www.senecapolytechnic.ca/about/policies/student-progression-and-promotion-policy.html)
  Outlines the requirements for students to graduate from an Ontario College Credential or an honours bachelor degree program at Seneca.
- Grading Policy (http://www.senecapolytechnic.ca/about/policies/grading-policy.html)
  Defines the standards for assigning grades to recognize student achievement in a course and program.
- Academic Appeal Policy (https://www.senecapolytechnic.ca/about/policies/academic-appeal-policy.html)
  Provides students with a fair, timely and consistent process to appeal decisions that impact their academic standing or progression, when the grounds for an academic appeal are met on the basis of merit of work, personal bias or unfair treatment, course management and/or extenuating circumstances.
- Prior Learning Assessment and Recognition Policy (https://www.senecapolytechnic.ca/about/policies/prior-learning-assessment-and-recognition-policy.html)
  Establishes a framework for assessing and recognizing credit for student's prior learning.
- Transfer Credit Policy (https://www.senecapolytechnic.ca/about/policies/transfer-credit-policy.html)
  Provides a framework for the evaluation and granting of transfer credits.

Technical Requirements

The following checklists outline the technical requirements for all students starting and continuing at Seneca:

Hardware checklist

- a computer that runs on Windows 10 or the latest Mac OSX and has up to date virus protection software
  - Windows 10 ARM64 (https://support.microsoft.com/en-us/windows/windows-10-arm-based-pcs-faq-477f51df-2e3b-f68f-31b0-06f5e4f8ebb5) devices are **not recommended** as they will not allow you to install AppsAnywhere, GlobalProtect, VPN, MyApps or use Virtual Commons and other virtual machine apps
- high-speed broadband access (Cable or DSL) is highly recommended. Some programs or courses require more advanced systems. Please refer to the program information page for information on specialized requirements
- headphones or speaker and a microphone for in-class conversations and meetings with your professors
- a webcam (may be required for specific courses)
- individual courses may have additional hardware requirements

Software checklist

- a web browser, such as Safari, Firefox, MS Edge, Google Chrome. Please note: You may need to upgrade your web browser to access online learning tools
- various applications are available to all full-time Seneca students, including Microsoft Office 365, Adobe Creative Suite, and Trend Micro
- Adobe Creative Suite includes a number of applications such as Premiere, Photoshop and more
- online teaching tools, including Blackboard, MS Teams, Zoom, BigBlueButton, and Webex
- individual courses may have additional software requirements for playing audio or video or other applications. You can also review the list of applications made available for home use on a Windows-based machine (http://myapps.senecapolytechnic.ca/)
  Note: Some applications may require you to install Student VPN to access licensed software
- antimalware software must be installed on all personal devices that will be used with your Seneca account. Visit the Malware and Virus Protection (https://students.senecapolytechnic.ca/spaces/185/it-security/wiki/view/963/malware-and-virus-protection) page for free and paid antimalware software recommendations, or visit the Trend Micro Internet Security (https://students.senecapolytechnic.ca/spaces/189/software/wiki/view/1360/trend-micro-internet-security) page for a free one-year license of this commercial antimalware software

Mobile devices checklist

- Mobile devices may allow for some participation in your course(s), however they present limitations and we cannot guarantee your device will meet all your coursework needs.
- All students are required to install and use Microsoft Authenticator (https://students.senecapolytechnic.ca/spaces/186/it-services/wiki/view/4168/microsoft-multi-factor-authentication) to access various services at Seneca. It's an important measure that provides an added layer of security on top of the login credentials for devices. In addition to using your username and password to log into these secure services, a second factor of authentication is required so that if your password becomes compromised, the intruder will not be able to log in. Use of multi-factor authentication is currently required for Blackboard, Office 365 (https://students.senecapolytechnic.ca/spaces/186/it-services/wiki/view/1003/office-365) and VPN (https://students.senecapolytechnic.ca/spaces/186/it-services/wiki/view/1024/vpn).
- A compatible Android (https://play.google.com/store/apps/details?id=com.azure.authenticator) or iOS (https://apps.apple.com/app/microsoft-authenticator/id983156458) mobile device that can be used to install Microsoft Authenticator is required.
- A cellphone data plan is not a mandatory requirement to use the Microsoft Authenticator app. The app can be used through a Wi-Fi connection or with no data connection.
- If you have a basic cellphone, you can choose to receive an SMS or a phone call as verification for second factor authentication.
- The Microsoft Authenticator app does not store any personal data.
- Authenticating through a mobile device is the only available option.

Helpful sites to bookmark:

- MySeneca.ca (https://outlook.office.com/mail/inbox) – access your Seneca email account
- Learn@Seneca (https://learn.senecapolytechnic.ca/ultra/institution-page) – Seneca's learning management system and intranet portal

Seneca Polytechnic Library Resources

Be sure to begin all research, assignment support and career preparation at Seneca Polytechnic Libraries (http://library.senecapolytechnic.ca) website. Students can find information about our services and collections including, print and e-books, databases that will lead to thousands of articles in magazines, newspapers, journals, encyclopedias, carefully selected websites, how-to tutorials, streamed videos and much more.

Citation Style Guidelines (https://library.senecapolytechnic.ca/citingsources): APA/MLA. Please check with your professor on the preferred formatting.

Seneca Policies

Below are some Seneca policies and links to more information.

Academic Integrity Policy (https://www.senecapolytechnic.ca/about/policies/academic-integrity-policy.html)

Seneca upholds a learning community that values academic integrity, honesty, fairness, trust, respect, responsibility and courage. These values enhance Seneca's commitment to deliver high-quality education and teaching excellence, while supporting a positive, equitable and inclusive learning environment. Review section 3 of the policy for details regarding approaches to supporting integrity. Section 4 and Appendix B of the policy describe various sanctions that can be applied, if there is suspected academic misconduct (e.g., contract cheating, cheating, falsification, impersonation or plagiarism).

Please visit the Academic Integrity at Seneca (http://open2.senecac.on.ca/sites/academic-integrity/for-students) website to understand and learn more about how to prepare and submit work so that it supports academic integrity, and to avoid academic misconduct.

Accessible Learning Services Policy (https://www.senecapolytechnic.ca/about/policies/accessible-learning-services-policy.html)

Seneca provides reasonable accommodations to students with disabilities to support academic success. If you require accommodation, contact the Accessible Learning Services Office *(*senecacnas@senecapolytechnic.ca*)* to initiate the process for documenting, assessing and implementing your individual accommodation supports for the classroom and Work-Integrated Learning (WIL) environments.

Accommodated students are required to meet the expected learning outcomes of courses. Accommodations do not surpass the need for safety or supersede academic policies and requirements.

Generative Artificial Intelligence (GenAI) Policy (https://www.senecapolytechnic.ca/about/policies/generative-ai-policy.html)

Seneca is committed to leveraging available technologies to work smarter, better and more efficiently. This includes the use of Generative Artificial Intelligence (GenAI) technologies. GenAI Frequently Asked Questions are available for students (https://students.senecapolytechnic.ca/spaces/288/copilot-seneca/wiki/view/15227/frequently-asked-questions-faqs). Violations of this policy may be subject to discipline.

Inclusion Policy (https://www.senecapolytechnic.ca/about/policies/inclusion-policy.html)

Every Seneca community member has the right to study and work in an environment that is free from discrimination, harassment, hate and racism. Individuals found to exhibit prohibited acts may face sanctions in accordance with relevant Seneca policies Information and assistance is available by contacting the Student Conduct Office at student.conduct@senecapolytechnic.ca.

Recording Lectures and Educational Activities Policy (https://www.senecapolytechnic.ca/about/policies/recording-lectures-and-educational-activities-policy.html)

Seneca students and employees are required to follow the framework of this policy for the collection, use, disclosure, and disposal of recordings of educational activities. The policy reinforces Seneca's support of recordings as a means to promote student learning and success.

By attending synchronous classes, students are consenting to the collection and use of their personal information for the purposes of administering the class and associated coursework. To learn more about Seneca's privacy practices, review Seneca's Privacy Notice (https://www.senecapolytechnic.ca/privacy.html).

Student Assessment Policy (https://www.senecapolytechnic.ca/about/policies/student-assessment-policy.html)

Seneca's approach to student assessment is grounded in commitments to equity and accessibility for students. Student work in academic courses will be assessed and evaluated to provide ongoing and meaningful feedback about their progress and development.

General Boilerplate

No electronic or wireless devices are permitted during the tests, including smartphones and Smart Glasses. If you need them, you must use prescribed non-Smart glasses.

All lab reports must be submitted individually by each student, regardless of individual or group work during the lab practices.

All inserted screenshots in lab reports must be identified by including either the student's full name or login account name and the date/time when it was taken.

All late lab report submissions are subject to a minimum 10% deduction. The later your submission, the higher the grade deduction.

Remember that a Satisfactory (minimum) grade for every lab report is 50%. If less, you must redo and resubmit the lab report as soon as possible, as the corresponding late-submission deductions are applied.

Please refer to the Blackboard course platform for more details.

Last updated: September 8, 2026 at 4:07 p.m.